Listener Inferences About Argumentative Framing in Quantified Descriptions of Exam Outcomes

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1 Listener-side experiment: inferring argumentative speaker types

While previous experiments in this project focused on the speaker's perspective, this study shifts attention to the listener's side. We examine whether listeners can (1) detect the use of argumentative language, (2) distinguish between different argumentative strategies (low vs. informative vs. high framing), and (3) infer the speaker's communicative intentions from utterances in context. Experimental items were selected in a model-driven, Bayesian manner, using posterior predictions from our speaker-side production models to identify utterance–state pairs that are diagnostic of distinct speaker types. In a forced-choice task, participants judged which type of speaker was most likely to have produced a given utterance. The experiment is preregistered .

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1.1 Method

1.1.1 Participants

Participants (N = 240) were recruited online via Prolific. Eligibility criteria were age 18+, native English, Prolific approval rate above 95%, at least five previously completed studies, and no participation in earlier related experiments from this project. Participants were paid £1.50 for an approximately 7-minute study (about £13/hour).

1.1.2 Design

We investigated one within-participant factor Speaker Type with three levels: informative (*info*), low-framing argumentative (*low-arg*), and high-framing argumentative (*high-arg*). Each condition determines a specific pairing of a state s and an utterance u. Given a $\langle s, u \rangle$ pair, the associated condition establishes the ground truth for the expected (modal) listener response. For example, in the *high-arg* condition we expect that *high-arg* will be the modal choice.

1.1.3 Materials

Model-driven item selection. Items were constructed using a model-driven Bayesian procedure based on posterior predictions from prior production-modeling analyses. For each candidate utterance–

state pair (u, s), we computed a posterior distribution over speaker types via Bayes' rule:

$$p(\sigma \mid s, u, M, \theta) \propto p(u \mid s, M, \theta, \sigma)\, p(\sigma), \tag{1}$$

where $\sigma \in \{\textsc{low}, \textsc{info}, \textsc{high}\}$ denotes the speaker's type, u is an utterance, s is a state, M is a model class, and θ represents posterior draws of model parameters. We assumed a uniform prior over speaker types. For each model M, we first marginalized over θ to obtain a posterior distribution over speaker types conditional on each utterance–state pair $\langle s, u \rangle$.

Utterance–state pairs were ranked by a discriminability score targeting a speaker type of interest:

$$\Delta(u, s) = P(\sigma^* \mid u, s) - \sum_{\sigma \neq \sigma^*} P(\sigma \mid u, s), \tag{2}$$

where σ^* denotes the target speaker type. For each speaker type, we selected the top ten (u, s) pairs yielding the highest discriminability scores. Finally, we combined results across model classes using a leave-one-out (LOO)-based weighted model average, with weights derived from expected log predictive density (ELPD). The aggregated predictions were then passed through a softmax choice rule. Because the model-free model achieved the highest predictive accuracy in prior model comparisons, the LOO-weighted average is effectively dominated by its predictions .

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Manual review. After selecting items via the procedure above, we manually inspected all candidates to ensure their suitability for use in the experiment. Items that were linguistically unnatural or pragmatically odd were replaced by the next-ranked alternatives within the same category.

Lists. This procedure yielded 10 items per condition. Items were split evenly into two lists, such that each participant saw only half of the items in a single experimental run.

1.1.4 Procedure

The experiment was framed as a "Who said this?" identification task. Participants were introduced to a background story: a fictional school had recently administered several exams, and three groups of people had different communicative goals when describing the results. The exam results were displayed visually in the same format as in the production experiment. In the analysis, each role was mapped onto one of three labels: *high-arg*, *low-arg*, or *info*.

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- **Student** (low-argumentative framing; *low-arg*) wants the exam to sound *harder* (as if students had a low chance of success), because this makes her own performance appear better by comparison.
- **Teacher** (high-argumentative framing; *high-arg*) wants the exam to sound *easier* (as if students had a high chance of success), because this reflects positively on teaching quality and students' preparedness.

- **Examiner** (informative; *info*) aims to provide an *objective* and neutral description of the exam, without making it sound particularly hard or easy.

Before the main task, participants completed training trials analogous to those in the production experiment, extended here to cover all three speaker-type conditions. On each main trial, participants selected which of the three speaker types was most likely to have produced a given utterance, based on the reported exam outcome.

1.2 Results

1.2.1 Descriptive results

Figure 1 shows the proportion of each response type by condition, with bootstrapped 95% confidence intervals. In the *low-arg* condition, the modal response was *low-arg* (54.4%), followed by *info* (31.6%) and *high-arg* (14.0%). In the *high-arg* condition, the modal response was *high-arg* (61.2%), followed by *info* (21.6%) and *low-arg* (17.2%). In the *info* condition, responses were more evenly distributed: *high-arg* (40.8%), *info* (30.8%), and *low-arg* (28.4%).

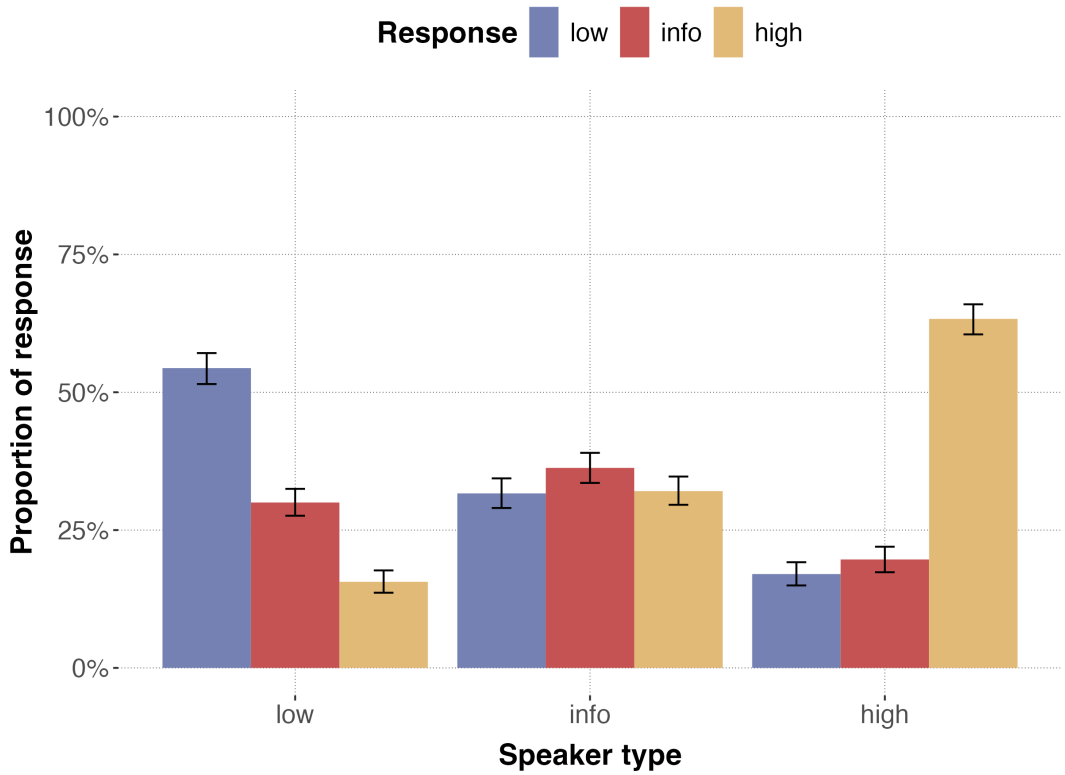

Figure 1: Proportion of response types by speaker-type condition, with bootstrapped 95% confidence intervals.

1.2.2 Bayesian regression

As preregistered, we fit a Bayesian hierarchical multinomial regression model with by-participant and by-item random intercepts and slopes, and used posterior expected predictions for hypothesis testing.

Above-chance modal selection. We first tested whether the proportion of modal responses exceeded chance level (1/3) in each condition. The posterior probability of above-chance performance was $P = 1.000$ for the *low-arg* condition and $P = 1.000$ for the *high-arg* condition. For the *info* condition, the posterior probability was only $P = 0.722$, falling short of the preregistered threshold of 0.95.

Conjunctive modal hypothesis. We then evaluated the conjunctive hypothesis that the *match* response was the modal choice in each condition, i.e., that it was more probable than *both* alternatives simultaneously. The posterior probability of this conjunctive hypothesis was $P = 1.000$ for the *low-arg* condition and $P = 0.995$ for the *high-arg* condition, both exceeding the preregistered threshold. For the *info* condition, the posterior probability was $P < 0.001$, indicating that listeners did not reliably identify the informative speaker as the modal choice.

Focused contrast: argumentative poles. As a more focused analysis, we compared the modal response against its direct opposite (e.g., *low-arg* vs. *high-arg*) in the *low-arg* and *high-arg* conditions. The posterior probability that the modal response was chosen over its opposite was $P = 1.000$ for the *low-arg* condition and $P = 0.995$ for the *high-arg* condition, confirming that listeners reliably distinguished the two argumentative poles.

1.3 Discussion (placeholder)

If the primary target is supported, the results suggest that listeners are sensitive not only to truth-conditional compatibility but also to the argumentative "direction" implied by quantified language. If it is not supported, the results would motivate refinements to the linking assumptions between production models and listener inferences (e.g., representations of speaker goals, salience of role descriptions, and the aggregation used in item selection).

References